Ramasamy Rajendran

Backend and AI Infrastructure Engineer

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SUMMARY

Backend and AI Infrastructure Engineer with 4+ years of experience building production-grade AI platforms from the ground up. At Labelbees, designed and owned the full backend stack spanning model inference, data pipelines, authentication, and API management for a multi-tenant enterprise AI platform on Kubernetes. Proficient in Python, FastAPI, and distributed systems, with hands-on experience integrating OSS and commercial AI models (CLIP, GPT-4, Gemini, Claude) into production workflows. Shipped scalable services handling 30K+ daily requests with 75% lower infrastructure costs through event-driven architecture and autoscaling.

TECHNICAL SKILLS

Languages: Python
Frameworks: FastAPI, SQLAlchemy, Celery, Pydantic, Asyncio
Architecture & Design: Microservices, REST API, Distributed Systems, Event-Driven Architecture, System Design, Async Programming, Data Modeling, Auth Systems
Cloud & Infrastructure: AWS, Kubernetes, Docker, Argo Workflows, KEDA
Databases: PostgreSQL (pgvector), Redis
ML & Computer Vision: PyTorch, ONNX, SageMaker, Triton
Geospatial: GDAL, Rasterio, GeoPandas, Map Servers
Gateway & Security: Istio, Envoy, Keycloak, JWT, API Keys, RBAC, Multi-tenancy
Observability: Prometheus, Grafana, CloudWatch, OpenTelemetry, Structlog
Methodologies: Agile/Scrum
Developer Tools: Linux, Git, Jupyter, Postman, JIRA, Claude Code

EXPERIENCE

Software Engineer Mar. 2022 – May 2026
Labelbees *India*

- Architected a multimodal AI platform to transform raw data into training-ready datasets, engineering 6 microservices and 2 data pipelines on GPU-backed Kubernetes pods, scaling the system to handle **30K+ daily requests** across real-time and batch workloads.
- Built the inference engine supporting near real-time (event-driven) and batch (Argo Workflows DAG) execution modes; integrated 7+ OSS and enterprise models (**CLIP, Grounding DINO, GPT-4, Gemini, Claude**) to power detection, segmentation, captioning, and OCR workloads.
- Cut infrastructure costs by **75%** and sustained **5K peak concurrent requests** by implementing **KEDA event-driven autoscaling** to dynamically scale Kubernetes GPU worker pods from 0→N based on SQS queue depth.
- Enabled **sub-second similarity search** across **500K+ images** by building an event-driven CLIP embedding pipeline using S3, SQS, Lambda, and SageMaker, with results indexed in PostgreSQL **pgvector**.
- Built a data management API enabling organizations to bulk-upload datasets, connect cloud storage (S3, GCS), and search via NLP and vector similarity to feed model inference and human QA review workflows.
- Built a schema-driven **BYOK credential vault** using AWS Secrets Manager, letting users connect their own AI providers (OpenAI, Anthropic, Google) and cloud storage (S3, GCS) with automatic fallback to platform keys on quota exhaustion, keeping inference uninterrupted.
- Saved **$14,400+/year** over Auth0/Okta by deploying self-hosted OSS **Keycloak** as centralized IAM, supporting multi-tenant user management (CIAM), RBAC, and organization administration across the platform.
- Eliminated per-service auth boilerplate across 7 microservices by centralizing authentication, rate limiting, and routing through **Istio gateway** with a custom authz service, enforcing JWT/API key validation, IP-based DDoS protection, and tier-based limits via Redis.

Process Executive Jul. 2019 – Sep. 2021
Mahendra Next Wealth IT India Pvt. Ltd. *India*

- Managed high-volume Flipkart customer escalations across 90+ brand clients via CRM and client email communication; trained and supervised a 15-member team to consistently resolve first-level escalations within 72-hour SLAs, including time-critical social media cases within 24 hours.
- Cut file splitting and format conversion time from 2+ hours to under 10 minutes (94% reduction) by building an Excel VBA automation and training 50+ team members to adopt it across all brands.
- Eliminated manual report generation by building a Python reporting system with pandas and numpy, cutting turnaround from 2 hours to 3 seconds (99.9% reduction).

EDUCATION

Periyar University Salem, India
Bachelor of Computer Application (BCA) — CGPA: 7.13/10 *2016 – 2019*